Project Proposal CS 546-B

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Project Description

Our project is a web application that helps people find housing in New York City more easily and transparently. The app will display available residential buildings and apartments, including important details like location, rent-stability status, building size, and other information that is publicly available. The app will provide the opportunity for users to leave a rating and review on each page. The application will also allow users to bookmark, favorite, and keep track of certain buildings that either interest them or to avoid them. Through combining the official NYC open house data with user-focused features, the application creates a dynamic and useful way for renters to explore their housing options and make informed decisions when searching for a place to live long term or short term.

Core Features

I. Landing Page
    A. Explains to the user the purpose and functionality of our platform.
    B. Basic instructions on how to navigate the website
    C. At the bottom of the landing page, include a survey on the quality of the application
II. User Profile Page
    A. Shows the User's Name
    B. Sign out Button
    C. Email <Can be changed on the page>
    D. Phone Number <Also can be changed on the page>
    E. User's Address (Maybe?...)
III. Buildings Page
    A. List view of buildings
    B. A search feature
IV. Building Page (a more detailed page for a given building selected by the user via the buildings page)
    A. Full building information: address, borough, block/lot, number of units, rent-stabilization status
    B. Composite Trust Score calculated from: number of NYC Housing Maintenance Code Violations, rent stabilization status, and average user review rating
    C. Violation history pulled from the NYC Housing Maintenance Code Violations dataset, displayed in a sortable table

    D. User reviews section: each user can leave one review with a star rating (1–5), and can edit or delete their own review
    E. User-reported issues section: users can report specific problems (e.g., pests, noise, maintenance neglect) separate from official violations
    F. Favorite/bookmark button to save the building to their profile- list can be found on buildings page with favorites filter
    G. Comments: Multiple comments per user allowed, with timestamps and delete functionality, clearly distinguished from the one-review-per-user system

V. Review System
    A. Each logged-in user can leave exactly one review per building
    B. Reviews include a star rating (1–5) and written feedback
    C. Users can edit or delete their own review
    D. Average rating is displayed on both the building card and detail page
    E. Reviews are sorted by most recent by default

VI. Admin Dashboard:
    A. Pre-seeded account, can edit/add/delete any building (including dataset-sourced ones), can moderate reviews/comments, has a dedicated dashboard

Extra Features

I. Direct messaging system with the landlord/agent

II. Map
    A. Interactive map (using Leaflet.js or Google Maps SDK) showing building locations as pins
    B. Clicking a pin navigates to that building's detail page
    C. Map filters match the list view filters (borough, rent-stabilized, violation count)

III. Notifications
    A. Set specific types of notifications
        1. New listing
        2. Update to an existing listing
        3. A direct message from a landlord/agent

IV. Recently viewed buildings
    A. A list of recently visited buildings

V. Notifications
    A. Users can opt in for:
        a. New listings in a saved borough or neighborhood
        b. Updates to a favorited building (new violations, new reviews)

VI. Question and Answers:
    A. Q&A on building pages - users can post questions about a building ("Is this place pet friendly?" or “Is it easy to hear the neighbors?”) and other users or even verified tenants can answer
    B. Upvote/downvote reviews, users can mark a review as helpful or not
    C. Comments on reviews so users can reply to each other's reviews

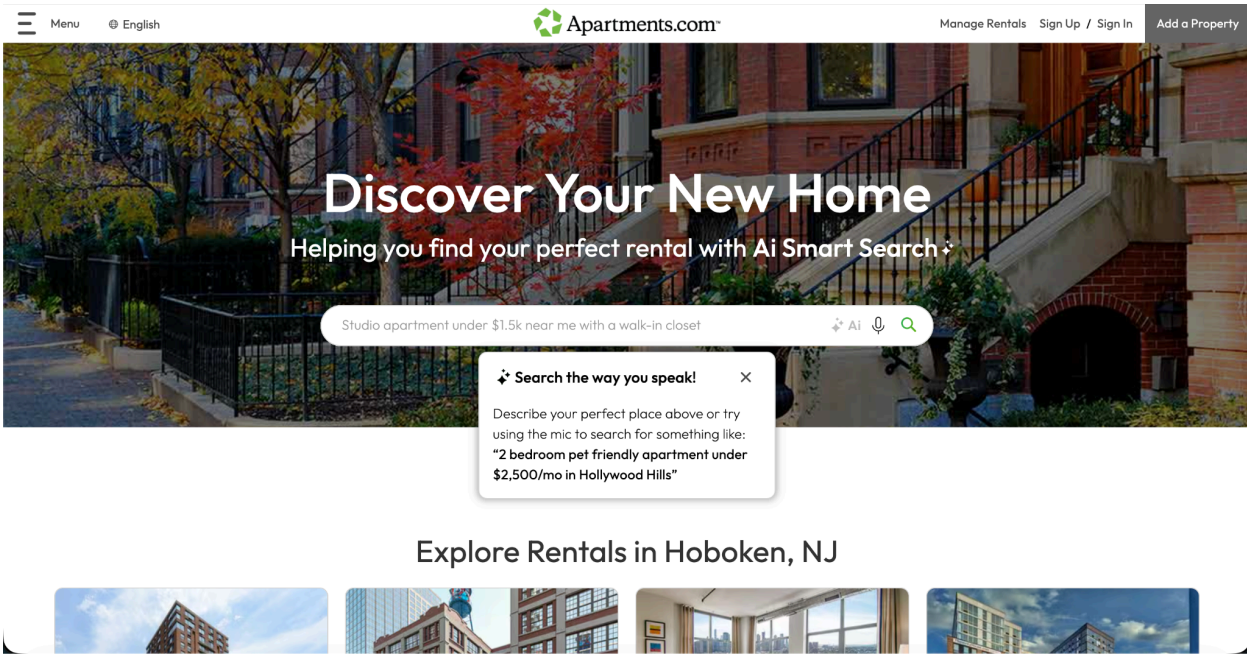
VII. Other:
    A. A visual Price Graph (a trend line) of an apartment/home.

B. Violation severity explainer so instead of just showing raw violation codes, translate them into plain English with a severity rating (minor / serious / hazardous) pulled from NYC violation class data
C. Moving checklist generator so before or after favoriting a building, users can generate a printable/downloadable moving checklist

Potential Datasets
Housing Database:
https://data.cityofnewyork.us/Housing-Development/Housing-Database/6umk-irkx/about_data
Housing Maintenance Code Violations:
https://data.cityofnewyork.us/Housing-Development/Housing-Maintenance-Code-Violations/wvxf-dwi5/about_data
Rent-stabilized buildings pdf dataset:
https://rentguidelinesboard.cityofnewyork.us/resources/rent-stabilized-building-lists/
Github:
GitHub URL here
https://github.com/livictor271/StableStay

Project Plans

1. Landing Page
    a. Register/sign in
    b. Contact Support
    c. Pictures
    d. About
    e. Menu (buttons to Buildings Pg and Profile Pg)

f. 

   i. reference^

2. User Profile: Could just be a button or included in the menu drop down.
3. Buildings Page:
   a. Filter
   b. Favorites
   c. City, Zip Code
   d.